The 2026 Ollama Deployment Matrix: Frontier Intelligence on Distributed and Local Systems

The state of local artificial intelligence in 2026 represents the fruition of several years of intensive research into architectural efficiency, sparse scaling, and the democratization of reasoning capabilities. The Ollama ecosystem has evolved from a simplified CLI tool into a sophisticated orchestration layer capable of managing multi-modal models that rival proprietary cloud-based systems. By early 2026, the distinction between "local" and "frontier" intelligence has largely dissolved, replaced by a nuanced matrix of deployment options where parameter count is often less important than the architectural paradigm—be it dense, Mixture-of-Experts (MoE), or reasoning-distilled weights. This report provides an exhaustive analysis of the 2026 Ollama deployment landscape, examining the specific hardware requirements, comparative performance benchmarks, and the architectural innovations that define the current era of decentralized compute.

The Architectural Evolution of the 2026 Model Library

The transition from the dense architectures prevalent in 2023 and 2024 to the sparse Mixture-of-Experts and reasoning-optimized models of 2026 has fundamentally altered the hardware calculus for local deployment. In the 2026 model library, models are no longer judged solely by their total parameter counts, but by their activated parameter counts and the efficiency of their attention mechanisms.[1] The Llama 4 family, the Qwen 3.5 series, and the DeepSeek-R1 updates represent the three pillars of this new intelligence economy.

Meta Llama 4: The Herd of Multimodal Sparse Experts

Meta's release of the Llama 4 family in early 2025 set the benchmark for open-weight multimodal performance. Moving beyond the dense structures of Llama 3, the Llama 4 series—comprising Scout, Maverick, and the teacher-model Behemoth—utilizes a Mixture-of-Experts architecture designed for native multimodality and extreme context lengths.[2] The Scout variant, with 109 billion total parameters, activates only 17 billion per forward pass across 16 experts.[2] This design allows Scout to maintain high inference speeds while supporting an industry-leading 10-million-token context window, facilitated by Interleaved Rotary Positional Embeddings (iRoPE).[3]

Maverick, the mid-tier professional model, utilizes 128 experts to reach approximately 400 billion total parameters, yet it also maintains an activated count of 17 billion.[2] The higher expert density in Maverick allows for superior specialization in coding and complex reasoning tasks, often outperforming proprietary models like GPT-4o in head-to-head evaluations.[2] The

architectural shift toward "early fusion" is a defining characteristic of Llama 4, where visual and textual tokens are integrated within a single transformer backbone, allowing for deep cross-modal understanding that surpasses the "late fusion" methods of earlier vision models.[4]

Alibaba Qwen 3.5: Agentic Intelligence and Gated Efficiency

Alibaba's Qwen 3.5 series has established itself as the premier choice for agentic workflows, emphasizing terminal-based reasoning, UI navigation, and complex search.[10] The Qwen 3.5 architecture incorporates "Gated Delta Networks" and sparse MoE layers, delivering high-throughput inference with minimal latency.[11] By early 2026, Qwen 3.5-35B-A3B has become a standard for local developers, activating only 3 billion parameters per token while achieving performance levels comparable to Claude 3.5 Sonnet on coding benchmarks.[11]

The "Fast" and "Thinking" modes introduced in the Qwen 3.5-397B-A17B model exemplify the trend toward dual-path inference.[11] "Fast" mode allows for rapid, cost-efficient processing of routine instructions, while "Thinking" mode utilizes extended chain-of-thought processing for multi-step reasoning and professional knowledge work.[11] This model also boasts expanded multilingual support for 201 languages, making it a critical tool for global enterprise deployment.[12]

DeepSeek-R1-0528: The Democratization of Frontier Reasoning

DeepSeek's R1-0528 represents the state-of-the-art in reasoning models, utilizing Grouped Relative Policy Optimization (GRPO) to achieve performance parity with OpenAI's o1 and o3 series at a fraction of the compute cost.[15] The R1-0528 update, released in May 2025, resolved many of the stability issues found in earlier reasoning models, introducing native support for JSON output and function calling—features essential for local software agents.[16]

The most significant contribution of the DeepSeek project remains its distillation initiative. By distilling the reasoning patterns of the 671 billion parameter R1 model into smaller dense architectures like Llama 3.3-70B and Qwen 2.5-32B, DeepSeek enabled high-level reasoning on consumer-grade hardware.[16] This has created a "Reasoning Economy" where small, 7B to 32B models can solve complex mathematical and logical problems previously reserved for massive proprietary clusters.[16]

The 2026 Ollama Deployment Matrix: Hardware and Comparison

The following table serves as the definitive 2026 matrix for Ollama model selection, mapping local models to their closest proprietary equivalents while detailing the minimum VRAM required for stable 32k context inference using 4-bit (Q4_K_M) quantization.

Model Category	Open-Source (Ollama) Model	VRAM (32k Context)	Proprietary API Equivalent	Primary Advantage
Ultra-Reasoning	DeepSeek-R1-0528 (671B)	430GB+	OpenAI o3 / Gemini 3 Pro	98.1% MATH-500 score [16]
Frontier General	Qwen 3.5-397B-A17B	250GB+	GPT-5.2 / Claude 4 Opus	"Fast" and "Thinking" dual-path modes [11]
Advanced Agentic	Llama 4 Maverick (400B)	256GB+	GPT-5.1 / Claude 4 Sonnet	128 experts for specialized coding tasks [2]
Long-Context RAG	Llama 4 Scout (109B)	84GB+	Gemini 3 Pro (1M+)	10-million-token context window [2]
High-Perf Coding	Qwen 3.5-35B-A3B	26GB+	GPT-4o / Claude 3.5 Sonnet	78.6 BrowseComp (Agentic Search) [11]
Desktop Reasoning	R1-Distill-Llama-70B	51GB+	OpenAI o1-mini	Frontier math on multi-GPU consumer builds [16]
Mobile/Edge	Mistral Small 3.2 (24B)	16GB+	GPT-4o mini	Latency-optimized for 150 tokens/s [21]
Compact Reasoning	Phi-4-Reasoning (14B)	12GB+	GPT-4o mini (Reasoning)	Strong instruction following at small scale [10]

Understanding the VRAM Calculation Formula

To deploy these models effectively, system architects must calculate VRAM requirements using

a formula that accounts for weights, system overhead, and the dynamic growth of the KV cache. In 2026, the standard formula utilized by the Ollama community for NVIDIA hardware is:

$$VRAM = (P \times b_w) + (0.55 + 0.08 \times P) + \frac{B \times N \times 2 \times L \times \frac{d}{g} \times b_{kv}}{10^9}$$

The variables represent:

- P : Parameter count in billions.[24]
- b_w : Bytes per weight (e.g., ~0.57 for 4-bit Q4_K_M, 1.0 for 8-bit Q8_0).[24]
- 0.55 : Constant CUDA overhead in GB for cuBLAS and workspace.[24]
- $0.08 \times P$: Scratchpad memory for activations and temporary tensors.[24]
- B : Batch size (concurrency).[24]
- N : Context length in tokens.[24]
- L, d, g : Model-specific architectural constants (Layers, Hidden Dimension, GQA grouping).[24]
- b_{kv} : Bytes per Key-Value scalar (default 2.0 for FP16, can be reduced via KV cache quantization).[24]

The implications of this formula are particularly severe for long-context models. For a 70B model, every 1,000 tokens of context adds approximately 0.11 GB of VRAM overhead.[24] At a context length of 1 million tokens, the KV cache alone can exceed 100 GB, necessitating the use of KV cache quantization (OLLAMA_KV_CACHE_TYPE=q4_0) or architectures like Multi-Head Latent Attention (MLA) found in Qwen and DeepSeek models.[26]

Hardware Infrastructure for the 2026 Local Stack

The release of NVIDIA's RTX 50-series and AMD's Instinct MI300 series has redefined the limits of local AI.[30] The RTX 5090, with 32 GB of GDDR7 VRAM and 1.8 TB/s of bandwidth, has become the gold standard for high-performance desktop inference.[30] The 78% increase in memory bandwidth over the 4090 allows for significantly faster routing in MoE models, where throughput is often memory-bound rather than compute-bound.[30]

GPU Performance and Bandwidth Comparison

GPU Model	VRAM	Bandwidth	AI TOPS	Performance

				Context (7B-8B Models)
RTX 5090	32 GB	1,792 GB/s	3,352	213 tokens/s [24]
RTX 4090	24 GB	1,008 GB/s	1,321	128 tokens/s [24]
RTX 4060 Ti	16 GB	288 GB/s	~300	89 tokens/s [24]
RTX 3090 (Used)	24 GB	936 GB/s	~280	42 tokens/s (70B Q4) [34]

For multi-GPU configurations, the introduction of PCIe Gen 5 support in the 50-series provides the necessary headroom for high-speed inter-GPU communication.[33] A dual-RTX 5090 setup can achieve over 7,000 tokens/second on small models like Llama 3.2-1B, while providing the 64 GB of VRAM required to run 70B-class models with generous context windows.[30]

Apple Silicon remains the primary alternative for large-parameter models like the Llama 4 Scout. The M4 Ultra, with up to 256 GB of unified memory, can load the entire weights of a 400B model, albeit at lower token-generation speeds (10-30 tokens/s) than dedicated NVIDIA clusters.[6] For many researchers, this "capacity over speed" trade-off is preferable for high-memory tasks like repository-wide code refactoring.[6]

Optimization and Operational Best Practices in Ollama 2026

Deploying frontier models locally requires more than just powerful hardware; it requires a sophisticated understanding of Ollama's internal scheduler and environment configuration. By 2026, the ollama serve command supports several high-impact flags that manage the tradeoff between latency, concurrency, and memory stability.

Concurrency and Load Management

The environment variable OLLAMA_NUM_PARALLEL determines how many simultaneous requests a single model instance can handle.[25] While increasing this number improves aggregate throughput, it multiplies the VRAM required for the KV cache.[25] For high-concurrency environments, architects should utilize the following settings:

- OLLAMA_NUM_PARALLEL=4: The balanced default for consumer GPUs with 16GB+ VRAM.[38]
- OLLAMA_MAX_LOADED_MODELS=1: Force-allocates all VRAM to a single high-parameter

model to prevent fragmentation.[38]
- OLLAMA_MAX_QUEUE=2048: Prevents request rejection during reasoning bursts.[38]

KV Cache and Attention Optimization

For models with massive context windows, FlashAttention (enabled via OLLAMA_FLASH_ATTENTION=1) has become a mandatory optimization, reducing the memory footprint and improving retrieval quality for long-context inference.[26] Furthermore, the introduction of KV cache quantization allows for a dramatic reduction in memory use. By setting OLLAMA_KV_CACHE_TYPE=q8_0, users can halve the VRAM consumed by context tokens with negligible impact on reasoning accuracy, making 128k context windows viable on single 24GB GPUs.[25]

In models like Qwen 3.5-35B-A3B, the "Gated Attention" mechanism allows for a hybrid approach where linear attention heads handle large-scale context while traditional attention heads focus on high-precision tasks.[14] This architectural versatility requires updated versions of llama.cpp (v0.2.x or later) to ensure correct layer offloading and thread synchronization across many-core CPUs.[26]

The ollama launch Command: Coding and Agentic Tools

One of the most significant feature additions in 2026 is the ollama launch command, which streamlines the integration of local models with coding agents like Claude Code, OpenCode, and Codex.[12] This command eliminates the need for manual API key configuration or environment variable management, automatically setting up an Anthropic-compatible Messages API on localhost:11434.[36] This allows tools that were previously "cloud-locked" to operate entirely on-premise, ensuring data sovereignty for sensitive enterprise repositories.[1]

Direct Benchmarking: Local Ollama vs. Proprietary API Performance

The 2026 performance landscape shows that local models have reached "Frontier Parity" in mathematics and reasoning, though they still lag in extreme-scale software engineering tasks.

Reasoning and Mathematical Performance

Benchmark	GPT-5.2 Thinking	OpenAI o3	DeepSeek-R1-0528	Qwen 3.5-397B	R1-Distill-70B

AIME 2025	100%	100%	92.0%	88.5%	86.7%
MATH-500	99.4%	98.4%	98.1%	97.4%	94.5%
GPQA Diamond	92.4%	87.7%	78.4%	70.4%	71.5%
ARC-AGI-2	52.9%	45.1%	31.1%	29.8%	25.4%

Data synthesized from: [12]

GPT-5.2 and o3 remain the leaders in abstract reasoning (ARC-AGI-2), which measures a model's ability to solve novel problems not found in its training data.[43] However, DeepSeek-R1-0528 and the Qwen 3.5 family have effectively closed the gap in graduate-level science and competition math.[12]

Coding and Agentic Performance

Benchmark	Claude 4 Opus	GPT-5.2 Pro	DeepSeek-R1	Qwen 3.5-35B	Maverick 17B
SWE-bench Verified	80.9%	80.0%	49.2%	61.2%	66.8%
SWE-bench Pro	52.0%	55.6%	38.4%	38.7%	40.2%
LiveCodeBench	68.4%	69.1%	62.1%	63.4%	65.5%

BrowseComp	84.0%	75.6%	78.6%	78.6%	72.4%

Data synthesized from: [11]

For software engineering, Claude 4 Opus and GPT-5.2 remain superior at multi-file refactoring and handling complex GitHub issues (SWE-bench Verified).[44] Local models like the Llama 4 Maverick and Qwen 3.5-35B excel at "agentic search" and single-file coding tasks, where their low latency allows for more iterations in a given development window.[8]

Specialized Local Models: Embedding, Coding, and Vision

The 2026 matrix includes specialized models for retrieval-augmented generation (RAG) and document understanding, where open-source options now dominate the cost-performance curve.

Embedding Models for RAG Pipelines

The Qwen 3 Embedding series, building upon the Qwen 3 foundation, provides a comprehensive range of multilingual text embeddings.[10] The Qwen 3-8B Embedding model has overtaken OpenAI's text-embedding-3-large as the highest-performing open-weight embedding model, offering 32k context and Matryoshka support for flexible output dimensions.[51]

Embedding Model	MTEB Score (Multilingual)	Context Window	Dimensions	Best For
Qwen 3 8B	70.58	32k	4096	Enterprise RAG [20]
Gemini-embedding-001	68.32	32k	3072	Multilingual API [53]
Nomic-v2-MoE	62.4	32k	768	Edge/Mobile search [20]
BGE-M3	63.0	8k	1024	Hybrid retrieval

				20

Vision and Multimodal Understanding

Multimodality in Ollama is now powered by an "early fusion" engine that allows models to treat visual tokens with the same granularity as text.[12] Llama 4 Scout and Maverick support up to 8 images per prompt and have demonstrated the ability to reason about scene progression and video frames.[7]

In document recognition, Qwen 3.5-122B has broken the 90% threshold on the OmniDocBench v1.5 benchmark, outperforming GPT-5.2 and Claude Opus 4.5 in extracting structured data from complex tables and charts.[11] This makes it an ideal tool for automated financial analysis and invoice processing within the Ollama ecosystem.[11]

Future Outlook: Scaling Efficiency and Data Sovereignty

As we look toward the remainder of 2026 and into 2027, several trends are emerging that will further refine the Ollama Deployment Matrix. The "scaling wall" of human-generated data has pushed research into synthetic data generation and "test-time compute" scaling.[57] Models like o1 and DeepSeek-R1 have proven that investing compute at the inference stage (thinking time) can produce massive leaps in intelligence without requiring larger parameter counts.[17]

The democratization of these techniques through open-source distillation ensures that the gap between a $2,000 desktop and a multi-million dollar API cluster will continue to narrow. For enterprises, the shift toward local deployment is no longer merely a cost-saving measure but a strategic necessity for maintaining data sovereignty in an era of agentic automation.[1] The Ollama ecosystem, by providing the bridge between cutting-edge research and accessible hardware, remains the definitive platform for this transformation.

The 2026 Ollama Deployment Matrix is thus a living infrastructure, where hardware selection (VRAM vs. Bandwidth), architectural choice (GQA vs. MLA), and operational tuning (FlashAttention and KV Quantization) converge to provide the most powerful AI experiences ever available to the individual user. Whether deploying a 1B model on a mobile device or a 671B reasoning engine on a local cluster, the capability to run frontier-level intelligence is now firmly in the hands of the community.

Works cited

1. Small Language Models for Phishing Website Detection: Cost, Performance, and Privacy Trade-Offs - MDPI, accessed March 9, 2026, https://www.mdpi.com/2624-800X/6/2/48
2. Rise of Multimodal LLMs: Benchmarking Models - Aisera, accessed March 9, 2026,

https://aisera.com/blog/multimodal-llm-llama4/
3. The Llama 4 herd | Hacker News, accessed March 9, 2026, https://news.ycombinator.com/item?id=43595585
4. llama-4-scout-17b-16e-instruct Model by Meta - NVIDIA NIM APIs, accessed March 9, 2026, https://build.nvidia.com/meta/llama-4-scout-17b-16e-instruct/modelcard
5. unsloth/Llama-4-Scout-17B-16E-Instruct-GGUF - Hugging Face, accessed March 9, 2026, https://huggingface.co/unsloth/Llama-4-Scout-17B-16E-Instruct-GGUF
6. Llama 4 Scout: Specifications and GPU VRAM Requirements - ApX Machine Learning, accessed March 9, 2026, https://apxml.com/models/llama-4-scout
7. Llama 4: Models, Architecture, Benchmarks & More | by Jatin Garg | Medium, accessed March 9, 2026, https://medium.com/@jatingargiitk/llama-4-models-architecture-benchmarks-more-4f297d6dc0fb
8. Llama 4 Models: Meta AI is Open Sourcing the Best - Analytics Vidhya, accessed March 9, 2026, https://www.analyticsvidhya.com/blog/2025/04/meta-llama-4/
9. Llama 4 Behemoth: Specifications and GPU VRAM Requirements - ApX Machine Learning, accessed March 9, 2026, https://apxml.com/models/llama-4-behemoth
10. library - Ollama, accessed March 9, 2026, https://ollama.com/library?sort=newest
11. Qwen3.5: Features, Access, and Benchmarks | DataCamp, accessed March 9, 2026, https://www.datacamp.com/blog/qwen3-5
12. qwen3.5 - Ollama, accessed March 9, 2026, https://ollama.com/library/qwen3.5
13. Alibaba's open Qwen 3.5 takes aim at GPT-5 mini and Claude Sonnet 4.5 at a fraction of the cost, accessed March 9, 2026, https://the-decoder.com/alibabas-open-qwen-3-5-takes-aim-at-gpt-5-mini-and-claude-sonnet-4-5-at-a-fraction-of-the-cost/
14. Qwen/Qwen3.5-35B-A3B - Hugging Face, accessed March 9, 2026, https://huggingface.co/Qwen/Qwen3.5-35B-A3B
15. OpenAI o3 vs DeepSeek r1: An Analysis of Reasoning Models - PromptLayer Blog, accessed March 9, 2026, https://blog.promptlayer.com/openai-o3-vs-deepseek-r1-an-analysis-of-reasoning-models/
16. DeepSeek-R1: Features, o1 Comparison, Distilled Models & More ..., accessed March 9, 2026, https://www.datacamp.com/blog/deepseek-r1
17. Reasoning Model Comparison Guide: DeepSeek R1 vs o3 vs Gemini 3 | Meta Intelligence, accessed March 9, 2026, https://www.meta-intelligence.tech/en/insight-reasoning-models
18. deepseek-r1 - Ollama, accessed March 9, 2026, https://ollama.com/library/deepseek-r1
19. R1 0528 vs o3 Deep Research (Comparative Analysis) - Galaxy.ai Blog, accessed March 9, 2026, https://blog.galaxy.ai/compare/deepseek-r1-0528-vs-o3-deep-research
20. Best Open-Source LLMs for RAG in 2026: 10 Models Ranked by Retrieval Accuracy, accessed March 9, 2026, https://blog.premai.io/best-open-source-llms-for-rag-in-2026-10-models-ranke

d-by-retrieval-accuracy/
21. Mistral Small 3, accessed March 9, 2026, https://mistral.ai/news/mistral-small-3
22. Mistral Small 3 vs Mistral Small 3.2 24B (Comparative Analysis) - Galaxy.ai Blog, accessed March 9, 2026, https://blog.galaxy.ai/compare/mistral-small-24b-instruct-2501-vs-mistral-small-3-2-24b-instruct
23. library - Ollama, accessed March 9, 2026, https://ollama.com/library?sort=popular%E0%A4%AA%E0%A4%B0
24. VRAM Calculator for Local Open Source LLMs - Accurate Memory ..., accessed March 9, 2026, https://localllm.in/blog/interactive-vram-calculator
25. FAQ - Ollama's documentation, accessed March 9, 2026, https://docs.ollama.com/faq
26. Optimizing Ollama Performance on Windows: Hardware, Quantization, Parallelism & More | by Kapil Khatik | Jan, 2026 | Medium, accessed March 9, 2026, https://medium.com/@kapildevkhatik2/optimizing-ollama-performance-on-windows-hardware-quantization-parallelism-more-fac04802288e
27. MiniCPM-SALA: Hybridizing Sparse and Linear Attention for Efficient Long-Context Modeling - arXiv, accessed March 9, 2026, https://arxiv.org/html/2602.11761v2
28. DeepSeek Innovation: How it Beats GPT ? Multi-Head Latent Attention (MLA) Part-1, accessed March 9, 2026, https://pub.towardsai.net/deepseek-innovation-how-it-beats-gpt-multi-head-latent-attention-mla-part-1-a66dd18232b9
29. The Attention Arms Race: How Modern Open-Source LLMs Are Reinventing the Transformer's Core - Maxim AI, accessed March 9, 2026, https://www.getmaxim.ai/blog/the-attention-arms-race-how-modern-open-source-llms-are-reinventing-the-transformers-core/
30. RTX 5090 vs 4090: Upgrade Guide for AI and Creators - Fluence Network, accessed March 9, 2026, https://www.fluence.network/blog/rtx-5090-vs-4090/
31. Hardware support - Ollama's documentation, accessed March 9, 2026, https://docs.ollama.com/gpu
32. The Definitive Guide to Local LLMs in 2026: Privacy, Tools, & Hardware - SitePoint, accessed March 9, 2026, https://www.sitepoint.com/definitive-guide-local-llms-2026-privacy-tools-hardware/
33. RTX 4090 vs RTX 5090 vs RTX PRO 6000: Comprehensive LLM Inference Benchmark | CloudRift Blog, accessed March 9, 2026, https://www.cloudrift.ai/blog/benchmarking-rtx-gpus-for-llm-inference
34. 5 Best GPUs for AI 2026: RTX 5090 vs 4090 vs 3090 Tested | Local AI Master, accessed March 9, 2026, https://localaimaster.com/blog/best-gpus-for-ai-2025
35. Llama 4: How to Run & Fine-tune | Unsloth Documentation, accessed March 9, 2026, https://unsloth.ai/docs/models/tutorials/llama-4-how-to-run-and-fine-tune
36. Running Claude Code locally with Ollama and open-source models as a free alternative to the Anthropic API - GitHub Gist, accessed March 9, 2026, https://gist.github.com/AUAggy/ccf6df83c297e76191ff2de8eb6a5168

37. Best AI for Developers: Claude vs GPT vs Gemini Technical Comparison 2026 - Cosmic JS, accessed March 9, 2026, https://www.cosmicjs.com/blog/best-ai-for-developers-claude-vs-gpt-vs-gemini-technical-comparison-2026
38. is there parallelism inside Ollama? - Stack Overflow, accessed March 9, 2026, https://stackoverflow.com/questions/78188399/is-there-parallelism-inside-ollama
39. ollama parallelism - Reddit, accessed March 9, 2026, https://www.reddit.com/r/ollama/comments/1hmoaqf/ollama_parallelism/
40. Qwen3.5-35B-A3B quantization quality + speed benchmarks on RTX 5080 16GB (Q8_0 vs Q4_K_M vs UD-Q4_K_XL) - Reddit, accessed March 9, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rei65v/qwen3535ba3b_quantization_quality_speed/
41. Blog · Ollama, accessed March 9, 2026, https://ollama.com/blog
42. 13 Best Embedding Models in 2026: OpenAI vs Voyage AI vs Ollama | Complete Guide + Pricing & Performance, accessed March 9, 2026, https://elephas.app/blog/best-embedding-models
43. GPT-5.2 Benchmarks (Explained) - Vellum, accessed March 9, 2026, https://www.vellum.ai/blog/gpt-5-2-benchmarks
44. How GPT-5.2 stacks up against Gemini 3.0 and Claude Opus 4.5 - R&D World, accessed March 9, 2026, https://www.rdworldonline.com/how-gpt-5-2-stacks-up-against-gemini-3-0-and-claude-opus-4-5/
45. Gemini 3 Flash vs GPT-5.2 vs Claude Opus 4.5 vs Grok 4.1, The Real Winner Surprised Me | by Cogni Down Under | Medium, accessed March 9, 2026, https://medium.com/@cognidownunder/gemini-3-flash-vs-gpt-5-2-vs-claude-opus-4-5-vs-grok-4-1-the-real-winner-surprised-me-b43d0688452e
46. GPT-5.2 Benchmark Analysis: Performance Comparison vs GPT-5.1 & Gemini 3 Pro - Vertu, accessed March 9, 2026, https://vertu.com/ai-tools/gpt-5-2-benchmark-analysis-performance-compariso n-vs-gpt-5-1-gemini-3-pro/
47. AI Model Benchmarks Mar 2026 | Compare GPT-5, Claude 4.5, Gemini 2.5, Grok 4, accessed March 9, 2026, https://lmcouncil.ai/benchmarks
48. AI Code Review Gets Better When Models Debate: Claude vs Gemini vs Codex vs Qwen vs MiniMax - Milvus, accessed March 9, 2026, https://milvus.io/blog/ai-code-review-gets-better-when-models-debate-claude-vs-gemini-vs-codex-vs-qwen-vs-minimax.md
49. Best AI Model for Coding (2026): Swapping Models Changed Scores 1%. Swapping the Harness Changed Them 22%. - Morph, accessed March 9, 2026, https://www.morphllm.com/best-ai-model-for-coding
50. Embedding models - Ollama, accessed March 9, 2026, https://ollama.com/search?c=embedding
51. The Best Open-Source Embedding Models in 2026 - Bento, accessed March 9, 2026, https://www.bentoml.com/blog/a-guide-to-open-source-embedding-models
52. Qwen/Qwen3-Embedding-8B - Hugging Face, accessed March 9, 2026,

https://huggingface.co/Qwen/Qwen3-Embedding-8B
53. Best Embedding Models 2025: MTEB Scores & Leaderboard (Cohere, OpenAI, BGE), accessed March 9, 2026, https://app.ailog.fr/en/blog/guides/choosing-embedding-models
54. Top embedding models on the MTEB leaderboard - Modal, accessed March 9, 2026, https://modal.com/blog/mteb-leaderboard-article
55. Ollama's new engine for multimodal models, accessed March 9, 2026, https://ollama.com/blog/multimodal-models
56. Vision models - Ollama, accessed March 9, 2026, https://ollama.com/search?c=vision
57. (PDF) LLMOrbit: A Circular Taxonomy of Large Language Models -From Scaling Walls to Agentic AI Systems - ResearchGate, accessed March 9, 2026, https://www.researchgate.net/publication/399956620_LLMOrbit_A_Circular_Taxonomy_of_Large_Language_Models_-From_Scaling_Walls_to_Agentic_AI_Systems